

homework_4_#_3

Anshruta Rahar

2026-04-08

Busiest Airports Analysis

In this section I will be working with the busiest airports data set and presenting my analysis on the 6 busiest airports of my choosing across a set of years.

The 6 busiest airports that I chose for my analysis are:

- Hartsfield-Jackson Atlanta International Airport (ATL)
- Frankfurt Airport (FRA)
- Beijing Daxing International Airport (PKX)
- Indira Gandhi International Airport (DEL)
- Singapore Changi Airport (SIN)
- O'hare International Airport (ORD)

For my analysis I chose to work in Rstudio as I was working with a data set from Wikipedia and found RStudio to be much more useful in terms of wrangling that data and analyzing it in comparison to other languages.

In Rstudio, I primarily worked with three major packages: tidyverse, knitr, and rvest.

To start my analysis I first loaded the packages and the respective data set on the busiest airports from Wikipedia. Once I had loaded all the tables on the Wikipedia webpage, I then created 6 different data sets for each individual year ranging from 2020 to 2025 which only had data of the 6 airports i chose. After getting the 6 airport specific data sets for the 6 years I made a table adding all those separate data sets into one table which was then used for graphing the % change in the airports over the years of 2020-2025.

The Airport Data Table:

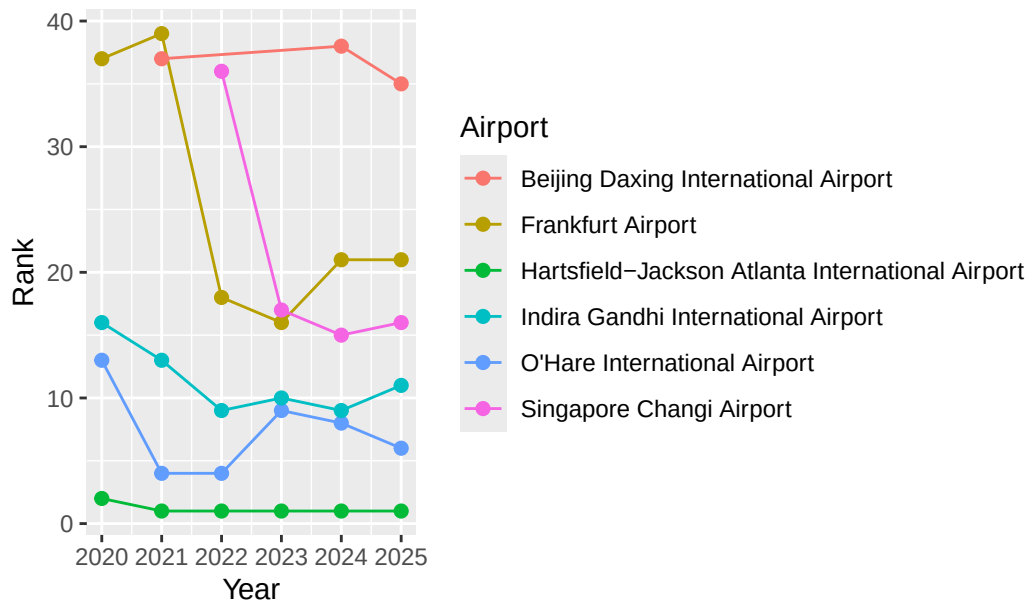
Table 1: Airport Data of the 6 selected airports from the year 2025 to 2020

Rank	Airport	Location	Country	Code(IATA/ICAO)	Total-passen- giers	Rankchange	%change	Year
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	106302208	NA	01.6%	2025
6	O’Hare International Airport	Chicago, Illinois	United States	ORD/KORD	851825	2	06.0%	2025
11	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	8000000	2	00.2%	2025
16	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	9980000	1	03.4%	2025
21	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	3189666	NA	02.6%	2025
35	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	3618949	3	08.4%	2025
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	108067766	NA	03.3%	2024
8	O’Hare International Airport	Chicago, Illinois	United States	ORD/KORD	8043050	1	08.3%	2024
9	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	7820834	1	07.8%	2024
15	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	7650000	2	14.8%	2024
21	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	3564957	5	03.7%	2024
38	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	441029	14	25.5%	2024
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	104653451	NA	11.7%	2023
9	O’Hare International Airport	Chicago, Illinois	United States	ORD/KORD	73894226	5	8.1%	2023
10	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	72214841	1	21.4%	2023
16	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	3355389	2	21.3%	2023

Rank	Airport	Location	Country	Code(IATA/ICAO)	Total-passenger-traffic	Rank change	% change	Year
17	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	\$8946000	19	83.1%	2023
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	\$3699630	NA	23.8%	2022
4	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	\$340619	NA	26.5%	2022
9	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	\$9490074	4	60.2%	2022
18	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	\$918482	21	97.2%	2022
36	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	\$2202000	NA	954.8%	2022
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	\$5704760	1	76.4%	2021
4	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	\$4020399	9	75.1%	2021
13	Indira Gandhi International Airport	Delhi	India	DEL/VIDP	\$7139957	3	30.3%	2021
37	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	\$5051012	16	55.7%	2021
39	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	\$4812849	2	32.2%	2021
2	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	\$42918685	1	61.2%	2020
13	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	\$3860251	7	63.5%	2020
16	Indira Gandhi International Airport	Delhi	India	DEL/VIDP	\$8500545	1	58.4%	2020
37	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	\$768601	22	73.4%	2020

The Airport Data Plot:

%Change in the airports over the years of 2020–2025



From the graph and table, I was able to form my analysis that Hartsfield-Jackson International Airport maintained its rank of being number 1 in comparison to the rest of the airports in the data sets and was consistent with its ranking of being number 1 after 2020 where its ranking was number 2 overall. O'hare then followed being the second highest ranked airport in 2020 and maintained its position of being second highest ranked airport in my choosing of the 6 respective airports even though its rankings were not as uniform yearly as they were for Hartsfield-Jackson and the other airports. Indira Gandhi then followed being the third highest ranked airport among the data set of the 6 chosen airports and maintained its position till 2025. Indira Gandhi's rank changes were also a little more uniform in comparison to O'hare and is the third airport which has uniform changes in rankings in comparison to the other airports with Jackson being the first and Beijing Daxing being the second airport with uniform ranking change throughout the years even though it's the lowest ranked airport overall. Frankfurt and Beijing had the same ranking at their starting years but Frankfurt seems to be the only one of the two which significantly improved its ranking and made itself the fifth highest ranked airport in my chosen list of the 6 airports. Singapore's Changi airport is the only airport which has improved its ranking the most among the 6 airports which is shown by the steepest decline in ranking from 2021 to 2022 and made itself the forth highest ranked airport among the 6 total airports.

Monte Carlo Numerical Integration

Initially I hadn't submitted any code for it in our previous homework as I couldn't choose a function so to keep things super simple I am choosing the function $f(x) = -(x-4)^2 + 4$ where

$2 \leq x \leq 6$ and output y in $0 \leq y \leq 4$.

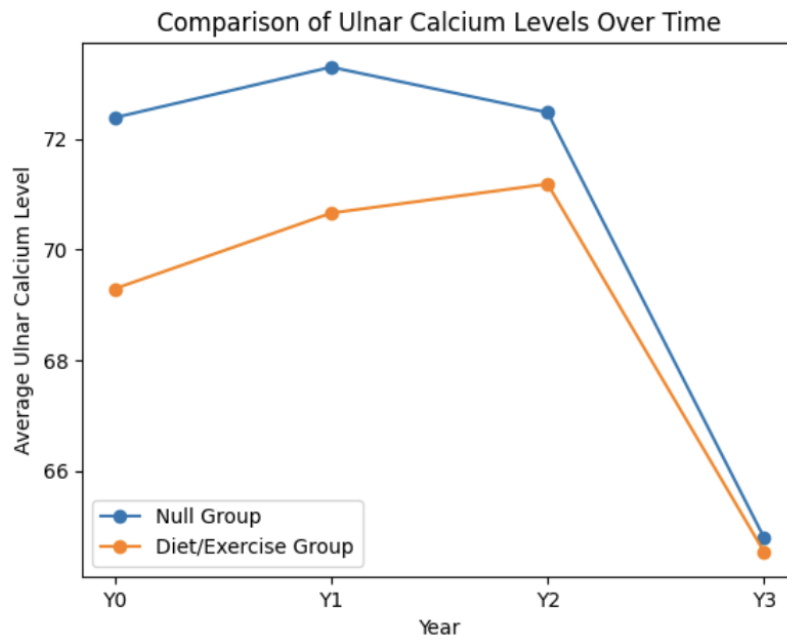
I am having a difficult time writing this simulation further but so far I started with loading my tidyverse package as I need to use if/else statement. Then I defined my function along with defining the bounds of both y and x . Once this was done I calculated the area of the rectangle as defined in the Question 21 of homework #3.4 and then moved on to creating/writing out the code for my Monte Carlo simulation where I randomly sampled n x and y coordinates using uniform distribution while following the instructions and steps laid out in question 21. Once they were joined together in a data frame I moved on to finding whether the ordered pair (x,y) were above or on/below the function.

Planning and Prompting Gen AI Tools

My Plan for the calcium data set:

1. The heading needs to be changed to something more descriptive so that the reader or first time viewer can know exactly or at least get a good idea of what the data set is trying to convey.
2. The first four columns are the data on the 15 women who were in the null group and the rest of the other four column are on the 16 women who were in the exercise/diet group. So I think to make it easier the data set should have two main columns where one is for null group and the second column is for the diet/exercise. The two columns then go into four other columns each for year0, year1, year2, year3 for both the two groups.
3. I now want to create a plot that shows the difference in the two groups of women in their ulnar calcium level throughout the four years.

The result I got from my prompt above:



You're thinking about this exactly the right way—your restructuring idea will make the dataset much clearer and easier to analyze.

1. Improving the structure (your plan refined)

Right now your dataset is “wide” and a bit hard to interpret. A cleaner structure would look like this conceptually:

Group	Year	Calcium
Null	0	...
Null	1	...
...	...	...
Diet/Exercise	0	...
...	...	...

This is called a *long format*, and it's ideal for plotting and statistical analysis. Your idea of grouping by **Null vs Diet/Exercise** and then tracking across **Year 0–3** is exactly right.

2. What your plot is showing

I created a plot of the **average ulnar calcium levels** for both groups across the four years (shown above).

Here's how to interpret it:

- Both groups **increase slightly from Year 0 → Year 2**
- The **diet/exercise group starts lower**, but narrows the gap over time
- At **Year 3**, both groups show a **sharp drop**
- The two groups are actually **very close by Year 3**

3. Key takeaway

Your visualization successfully highlights the **trend over time**, not just raw values:

- The intervention (diet/exercise) doesn't dramatically outperform the null group
- But it shows a **more stable trajectory through Years 1–2**
- The sharp drop in Year 3 for both groups is something worth questioning (measurement issue? biological effect?)

4. If you want to improve the plot further

A few ideas to make it stronger:

- Add **error bars** (to show variability)
- Plot **individual trajectories** (not just averages)
- Use a clearer title, for example:
"Average Ulnar Calcium Levels Over 4 Years: Null vs Diet/Exercise Groups"



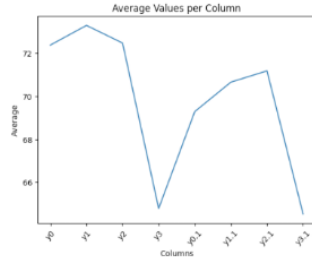
If you want, I can also help you:

- ↳ Reshape your dataset into the clean "long format"
- ↳ Run a statistical test (like repeated measures or t-tests)
- Or ↳ make a more polished, publication-quality graph `</>`



The generic result given to me in openMath:

Prompt: Create a data visualization for these data. Calcium.csv



I created a visualization showing the **average value of each column** in your dataset.

What this shows:

- Each point represents the **mean (average)** of one column (y0, y1, y2, y3, etc.).
- You can quickly compare which variables tend to have higher or lower values overall.
- For example, some columns cluster around ~70–73, while others drop closer to ~65.

If you want something more insightful, I can also create:

- Boxplots (to show spread and outliers)
- Time series plots (if rows represent time/order)
- Correlation heatmaps
- Side-by-side comparisons (e.g., y0 vs y0.1)

Just tell me 👍

Prompt: What is your version information?

You're chatting with **ChatGPT based on the GPT-5.3 architecture**.

Key details:

- **Model:** GPT-5.3
- **Knowledge cutoff:** August 2025
- **Current date (system):** March 31, 2026

If you need more technical details (capabilities, differences vs GPT-4/5, etc.), just let me know



Looking at the two results, I believe my prompt seemed to have produced a better result as you can actually see the difference between the two groups instead of how the data is being

visualized in the graph provided to me in open math as the prompt doesn't ask for the data to be tidied. I can also see that my prompt resulted in a lot more information as takeaway from the data set in comparison to the one generated by the generic prompt. Another thing I noticed is that my prompt resulted in more informative answers in comparison to the generic prompt as the Gen AI tool seems to not have the whole idea of what the data set contains as the original data set is not wrangled.

Reflection

When I think back to the person I was before I took this course and the one I am now, I feel nothing but appreciation because though at times the course seemed a little difficult and time consuming I feel like it made me more responsible, informative, and most importantly communicative as there were numerous times in class or my homework where I had to share my ideas and findings to my classmates and my professor which made me become more observant of what I was trying to present and how I was trying to put my idea and perspective across in the best possible way that could be easily understood. Another thing this course taught me was how when all my classmates and I participate in class discussions we all tend to learn a lot more than when we don't as during class discussions we tend to bounce ideas off of each other which has helped me in building my understanding of what we are learning about along with learning how to look at the same question with perspectives other than myself as during the time when we collaborate with our table mates we tend to discuss and understand each others' answers and use all those together to come to a singular answer that we can answer as a group along with answering some questions individually.

Gen AI Usage Appendix

I used ChatGPT model **GPT-5.3** and used it for part c of the assignment for the calcium prompt and planning. I used it on Wednesday April 15th and the prompts I used were:

1. My Plan for the calcium data set: The heading needs to be changed to something more descriptive so that the reader or first time viewer can know exactly or at least get a good idea of what the data set is trying to convey. The first four columns are the data on the 15 women who were in the null group and the rest of the other four column are on the 16 women who were in the exercise/diet group. So I think to make it easier the data set should have two main columns where one is for null group and the second column is for the diet/exercise. The two columns then go into four other columns each for year0, year1, year2, year3 for both the two groups. I now want to create a plot that shows the difference in the two groups of women in their ulnar calcium level throughout the four years.
2. Which version of chatpgt are you?

The response I got were:

1. You're thinking about this exactly the right way—your restructuring idea will make the dataset much clearer and easier to analyze.

1. Improving the structure (your plan refined)

Right now your dataset is “wide” and a bit hard to interpret. A cleaner structure would look like this conceptually:

Group	Year	Calcium
Null	0	...
Null	1	...
...	...	...
Diet/Exercise	0	...
...	...	...

This is called a *long format*, and it's ideal for plotting and statistical analysis. Your idea of grouping by **Null vs Diet/Exercise** and then tracking across **Year 0–3** is exactly right.

2. What your plot is showing

I created a plot of the **average ulnar calcium levels** for both groups across the four years (shown above). Here's how to interpret it:

- Both groups **increase slightly from Year 0 → Year 2**
- The **diet/exercise group starts lower**, but narrows the gap over time
- At **Year 3**, both groups show a **sharp drop**
- The two groups are actually **very close by Year 3**

3. Key takeaway

Your visualization successfully highlights the **trend over time**, not just raw values:

- The intervention (diet/exercise) doesn't dramatically outperform the null group
- But it shows a **more stable trajectory through Years 1–2**
- The sharp drop in Year 3 for both groups is something worth questioning (measurement issue? biological effect?)

4. If you want to improve the plot further

A few ideas to make it stronger:

- Add **error bars** (to show variability)
 - Plot **individual trajectories** (not just averages)
 - Use a clearer title, for example:
“Average Ulnar Calcium Levels Over 4 Years: Null vs Diet/Exercise Groups”
-

If you want, I can also help you:

- Reshape your dataset into the clean “long format”
- Run a statistical test (like repeated measures or t-tests)
- Or make a more polished, publication-quality graph

2. I’m based on **GPT-5.3**, one of OpenAI’s newer large language models.

If you’re comparing versions (like GPT-4, GPT-4o, etc.), I can also break down the differences depending on what you’re curious about.

Code Appendix

```
###Busiest Airport Code:

#Step1:loading packages
library(tidyverse)
library(rvest)
library(knitr)
library(ggplot2)

#Step2:load data from wikipedia
BusiestAirportRawData <- read_html(
  x = "https://en.wikipedia.org/wiki/List_of_busiest_airports_by_passenger_traffic"
) |>

  html_elements(css="table") |>
  html_table()

#Step3: creating a vector for the airports i want
```

```

airport_code <- c("ATL/KATL", "FRA/EDDF", "PKX/ZBAD", "DEL/VIDP", "SIN/WSSS", "ORD/KORD")

#Step4: loading the specific datasets of the specific years
busyAiportData2025 <- BusiestAirportRawData[[1]]
busyAiportData2024 <- BusiestAirportRawData[[2]]
busyAiportData2023 <- BusiestAirportRawData[[3]]
busyAiportData2022 <- BusiestAirportRawData[[4]]
busyAiportData2021 <- BusiestAirportRawData[[5]]
busyAiportData2020 <- BusiestAirportRawData[[6]]

#Step5: wrangling each dataset of each specific year
busyAiportData2025New <- busyAiportData2025 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2025, Totalpassengers = parse_number(Totalpassengers))

busyAiportData2024New <- busyAiportData2024 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2024, Totalpassengers = parse_number(Totalpassengers))

busyAiportData2023New <- busyAiportData2023 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2023, Totalpassengers = parse_number(Totalpassengers))

busyAiportData2022New <- busyAiportData2022 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2022, Totalpassengers = parse_number(Totalpassengers))

busyAiportData2021New <- busyAiportData2021 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2021, Totalpassengers = parse_number(Totalpassengers))

busyAiportData2020New <- busyAiportData2020 |>
  filter(`Code(IATA/ICAO)` %in% airport_code) |>
  mutate(Year = 2020, Totalpassengers = parse_number(Totalpassengers))

#generating the table for airport data of the 6 airports
busyAiportDataTable <- bind_rows(
  busyAiportData2025New,
  busyAiportData2024New,
  busyAiportData2023New,
  busyAiportData2022New,
  busyAiportData2021New,

```

```

    busyAiportData2020New
)

kable(
  busyAiportDataTable,
  caption = "Airport Data of the 6 selected airports from the year 2025 to 2020"
)

```

Table 3: Airport Data of the 6 selected airports from the year 2025 to 2020

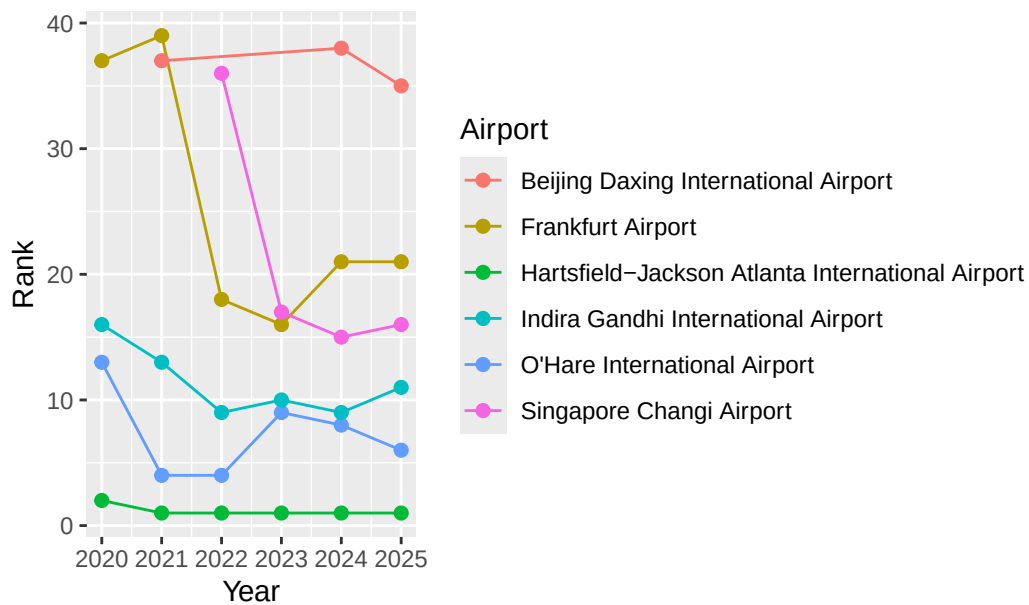
Rank	Airport	Location	Coun- try	Code(IATA/ICAO)	Total- passen- gers	Rank	change	Year
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	106302208	NA	01.6%	2025
6	O’Hare International Airport	Chicago, Illinois	United States	ORD/KORD	851825	2	06.0%	2025
11	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	8000000	2	00.2%	2025
16	Singapore Changi Airport	Changi, East Region	Singa- pore	SIN/WSSS	9980000	1	03.4%	2025
21	Frankfurt Airport	Frankfurt, Hesse	Ger- many	FRA/EDDF	8189666	NA	02.6%	2025
35	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	3618949	3	08.4%	2025
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	108067766	NA	03.3%	2024
8	O’Hare International Airport	Chicago, Illinois	United States	ORD/KORD	1043050	1	08.3%	2024
9	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	7820834	1	07.8%	2024
15	Singapore Changi Airport	Changi, East Region	Singa- pore	SIN/WSSS	87650000	2	14.8%	2024
21	Frankfurt Airport	Frankfurt, Hesse	Ger- many	FRA/EDDF	8564957	5	03.7%	2024
38	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	10441029	14	25.5%	2024

Rank	Airport	Location	Country	Code(IATA/ICAO)	Total-passenger-traffic	Rankchange	%change	Year
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	104653451	NA	11.7%	2023
9	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	73894226	5	8.1%	2023
10	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	72214841	1	21.4%	2023
16	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	69355389	2	21.3%	2023
17	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	58946000	19	83.1%	2023
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	93699630	NA	23.8%	2022
4	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	8340619	NA	26.5%	2022
9	Indira Gandhi International Airport	Palam, Delhi	India	DEL/VIDP	69490074	4	60.2%	2022
18	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	68918482	21	97.2%	2022
36	Singapore Changi Airport	Changi, East Region	Singapore	SIN/WSSS	32202000	NA	954.8%	2022
1	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	75704760	1	76.4%	2021
4	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	84020399	9	75.1%	2021
13	Indira Gandhi International Airport	Delhi	India	DEL/VIDP	7139957	3	30.3%	2021
37	Beijing Daxing International Airport	Daxing District, Beijing	China	PKX/ZBAA	45051012	16	55.7%	2021
39	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	64812849	2	32.2%	2021
2	Hartsfield–Jackson Atlanta International Airport	Atlanta, Georgia	United States	ATL/KATL	42918685	1	61.2%	2020
13	O'Hare International Airport	Chicago, Illinois	United States	ORD/KORD	8060251	7	63.5%	2020

Rank	Airport	Location	Country	Code(IATA/ICAO)	Total-passenger	Rankchange	%change	Year
16	Indira Gandhi International Airport	Delhi	India	DEL/VIDP	28500545	1	58.4%	2020
37	Frankfurt Airport	Frankfurt, Hesse	Germany	FRA/EDDF	18768601	22	73.4%	2020

```
#visualizing the airport data
busyAiportDataTable|>
  ggplot(
    mapping = aes(
      x = Year,
      y = `Rank`,
      color = Airport,
      group = Airport
    )
  ) +
  geom_line() +
  geom_point(size = 2)+
  labs(
    title = "%Change in the airports over the years of 2020-2025",
    x = "Year",
    y = "Rank"
  )
```

%Change in the airports over the years of 2020–2025



```
####Monte Carlo Code
#loading packages
library(tidyverse)

#defining the function
monte_function <- function(x) {
  -(x-4)^2 + 4
}

#calculating the area of rectangle
x_l <- 2
x_h <- 6
y_l <- 0
y_h <- 4

area_rect <- (x_h - x_l)*(y_h - y_l)

#monte carlo simulation
##randomly sampling n x and y-coordinates using uniform distribution
##calculating f(x)
##identifying whether each ordered pair (x,y) is above or below the func
monte_sim <- function(n, x_l, x_h, y_l, y_h){
```



```
r_sam <- data.frame(  
  x = runif(n, x_l, x_h),  
  y = runif(n, y_l, y_h)  
)  
  
r_sam$monte_function_x <- monte_function(r_sam$x)  
r_sam$check <- ifelse(r_sam$y<=r_sam$monte_function_x, "on/below", "above")  
}
```